U-I-500B
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SUPERSEDING
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March 8, 1972
Interim Revision of
Fed. Spec. U-I-500
November 8, 1960

FEDERAL SPECIFICATION

IMPRESSION MATERIAL, DENTAL
(Zinc-Oxide Eugenol Paste)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers both hard and soft dental impression paste for use in dental impression and bite registration procedures.

1.2 Classification. The impression materials covered by this specification shall be of the following types, as specified (see 6.1):

Type I - Hard
Type II - Soft

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issues in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein:

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Federal Specifications

U-S-746	Stone, Artificial, Dental.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Federal Standards

Fed. Std. 123	Marking for Domestic Shipment (Civilian Angencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Copies of listed federal and military standards, specifications, and Commercial Item Descriptions (CIDs) listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting activity.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

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(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN DENTAL ASSOCIATION (ADA)

ADA Spec. No. 16 for Dental Impression Paste - Zinc Oxide - Eugenol type.

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, Illinois 60611).

AMERICAN SOCIETY FOR TESTING AND MATERIALAS (ASTM)

ASTM D5 - Method of Test for Penetration of Bituminous Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila., Pa 19103).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

(Application for copies should be addressed to the Division of Public Information, Food and Drug Administration, Washington, DC 20201).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Material. The impression material shall consist of a zinc-oxide paste and a eugenol paste, each furnished in its own individual tube. The zinc-oxide paste (base) shall consist essentially of zinc oxide conforming to the tests, standards, and requirements of the USP, and other ingredients suitable for pharmaceutical or cosmetic use. The combined total eugenol content of the base paste and accelerator paste shall be no more than 15 percent eugenol for type I and no more than 19.5 percent eugenol for type II. The eugenol shall conform to the tests, standards, and requirements of the USP. Other ingredients in the eugenol paste shall be suitable for pharmaceutical or cosmetic use. The pastes shall be clean, smooth, homogeneous, and free from any grittiness or foreign matter.

3.2 Description and physical properties. Shall be a dental impression material paste consisting of a tube of base (zinc oxide), a tube of accelerator paste (eugenol), two keys for forcing material from tubes, and a set of instructions. Both pastes shall be extrudable from collapsible tubes and shall readily combine with each other when spatulated in the usual dental manner. The mixed paste shall be of such consistency as to record accurately the complete surface detail of the oral mucosa and associated soft tissues, and shall produce a smooth impression from which a satisfactory gypsum cast may be obtained. In addition, the soft impression material (type II) shall also be suitable for use as a bite registration medium. The resulting mixture shall be clean, homogeneous, free of foreign material, granules, and crystals. The mixed paste of type II shall be easily removed from skin or mucosa without the use of solvents. The mixed paste shall have the properties specified in Table I.

TABLE I. Physical property requirements.

Type	Consistency		Initial Setting		Final Setting	Penetration Hardness	
	Min.	Max.	Min.	Max.	Max.	Min.	Max.
	millimeters		minutes		minutes	millimeters	
I(Hard)	30	50	3-1/4	4-1/2	8	0.0	0.5
II(Soft)	20	45	3	6	15	0.8	1.5

3.2.1 Separation from cast. The material shall separate cleanly, with no evidence of adherence of the set paste to hardened artificial stone conforming to U-S-746, when tested as specified in 4.4.4.

3.2.2 Color. The zinc-oxide paste and the eugenol paste shall be distinctly different in color. The color of the combined pastes shall be pink, cream, or white and shall be uniformly dispersed. Any color additives used shall be approved and certified under the Federal Food, Drug, and Cosmetic Act.

3.2.3 Weight. The sum of the net weights of the zinc-oxide paste and the eugenol paste shall be for type I a minimum of 250 grams per unit issue and type II a minimum of 125 milliliters per unit issue. The two pastes shall be furnished in the relative proportions specified (see 3.4).

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3.3 Irritation. The impression material shall contain no ingredients, other than eugenol which are irritating to the oral tissues.

3.4 Instructions. Adequate and accurate instructions for proportioning and manipulating pastes shall accompany each unit package. Instructions shall include:

1. Proportioning of pastes both by weight and by extruded lengths.
2. Setting time.

Two copies of the instructions shall be forwarded to the contracting officer for information purposes.

3.5 Marking. Each tube, or as an alternate, each unit package, shall be marked with a lot, control, or batch number which shall refer to the manufacturer's records for the particular lots or batches. Each tube of paste shall be marked with the manufacturer's name or registered trademark. Tubes of base and accelerator shall be distinguished from each other by clearly marking the percent eugenol content on the tube of accelerator paste. Each tube of paste shall be marked with the generic name, or, as an alternate, the trade name of the item.

3.6 Immediate containers. Tubes for zinc-oxide paste and eugenol paste shall conform to PPP-C-186, Groups B, Class 1, Type a, Closure A. A total of two keys (one for each tube) shall be supplied for forcing the material from the tubes.

3.7 Certification. Impression paste shall be tested by the Council on Dental Materials and Devices for the purpose of establishing a list of certified materials. Suppliers shall furnish, at time of award, a dated certificate and data showing that the brand of impression paste they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification Number 16 for Dental Impression Paste-Zinc-Oxide Eugenol Type, type I - hard or type II - soft, as specified (see 6.1). Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

3.8 Workmanship. Impression paste shall be free from defects which detract from its appearance or impair its serviceability.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number and the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in table II and IV (see 4.2.1 and 4.3) and table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. Unit of product for examination purposes shall be the unit of issue (package of impression material).

Table II. Inspection and acceptable quality levels

Type Defect	Inspection Level	AQL (Percent Defective)
Major A	I	1.0
Major B	S-2	2.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with table III, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one unit of issue (package) taken at random from each lot or batch number. A lot shall be rejected if the average of the determinations for any unit of product fails the test.

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Table III. Tests

Characteristic	Requirement	Test procedure	Unit of Product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations per lot
Consistency	3.2	4.4.2	2	3	6
Initial setting time	3.2	4.4.3.1	2	3	6
Final setting time	3.2	4.4.3.2	2	3	6
Penetration hardness	3.2	4.4.3.3	2	3	6
Separation from Cast	3.2.1	4.4.4	1	2	2

4.3 Examination. Impression material shall be examined for defects, including but not limited to, those listed in table IV.

Table IV. Classification of defects

Categories	Defects
Major A:	
101	Type not as specified.
102	Color not as specified for base, accelerator, and mixed paste.
103	Presence of foreign material such as dirt, lint, or lumpy material. \ 1
104	Unit not complete with base and accelerator.
105	Manufacturer's instructions not present, do not include: a. Proportioning of pastes both by weight and by extruded length. b. Setting time.
Major B:	
151	Base and accelerator not packaged in tubes as specified.
152	Closure of tubes not as specified in 3.6.
153	Accelerator tubes not marked with eugenol content.
154	Caps not tight.
155	Holes and tears in collapsible tubes.
156	Liner not present in screwcap.
Minor:	
201	Tubes not commercially identified.
202	Labeling not complete or legible.
203	Two keys, for forcing material from tubes, not present in unit container.

\ 1 In process examination may be employed for these defects.

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4.4 Testing Procedures.

4.4.1 Preparation of test specimens. The preparation of test specimens shall be conducted at a temperature of 23.0 ± 2.0° C and at a relative humidity of 50 ± 10 percent. Equipment and materials shall be conditioned in the testing room for not less than 10 hours prior to making tests. The two pastes shall be measured by weight in the proportions specified by the manufacturer. The mixing techniques shall be that described by the manufacturer.

4.4.2 nsistency. One type of apparatus for measuring consistency is shown in figure 1. This apparatus consists of a glass tube (inside diameter approximately 10 mm) which will deliver a definite volume (0.5 ml) of mixed paste, two flat glass plates, a 500 gm weight and two thin plastic sheets. The combined weight of the top plate and the weight shall be 520 ± 2 gm. After mixing, 0.5 ml of the paste shall be placed in the measuring device and deposited on one thin plastic sheet on a glass plate. One and one-half minutes after starting the mix, a second plastic sheet, a second glass plate and the 500 gm weight shall be carefully placed on the soft paste. Ten minutes after starting the mix, the weight shall be removed and the major and minor diameters of the slumped mass of paste shall be measured to the nearest millimeter. The average value for three disks shall be reported to the nearest millimeter (The diameter of the disk is the measure of consistency).

4.4.3 Setting characteristics. A cup 25 mm in diameter and 3 mm deep shall be overfilled with impression paste prepared as described in 4.4.1 and the surface struck off level with a spatula. All times shall be measured from the time mixing is commenced.

4.4.3.1 Initial setting. A steel rod 2.4 mm in diameter with hemispherical ends and weighing 10 gm shall be applied to the surface of the material at 30-second intervals. The end of the rod shall be wiped clean after each application. The time of initial setting shall be the number of minutes elapsed from the starting of the mix to the time at which the material ceases to string when the rod is withdrawn. The time of initial setting shall be recorded to the nearest half minute. The value for time of initial setting shall be reported as the average of three determinations and shall be rounded off to the nearest half minute.

4.4.3.2 Final setting. The specimen used for measuring the time of initial setting shall be used for measuring the time of final setting. The cup shall be placed on the platen of a Krebs penetrometer (or similar instrument of equal accuracy) which shall be equipped with a cylindrical indenter 1.59 ± 0.013 mm in diameter, loaded so that the weight of the indenter and load shall be 50 gm. The indenter shall be applied to the surface of the specimen for 10 seconds at 30-second intervals and the penetration measured. The time of final setting shall be the number of minutes elapsed from the starting of the mix to the time when the penetration falls below 0.2 mm. The valve for time of final setting shall be reported as the average of three determinations and shall be rounded off to the nearest half minute.

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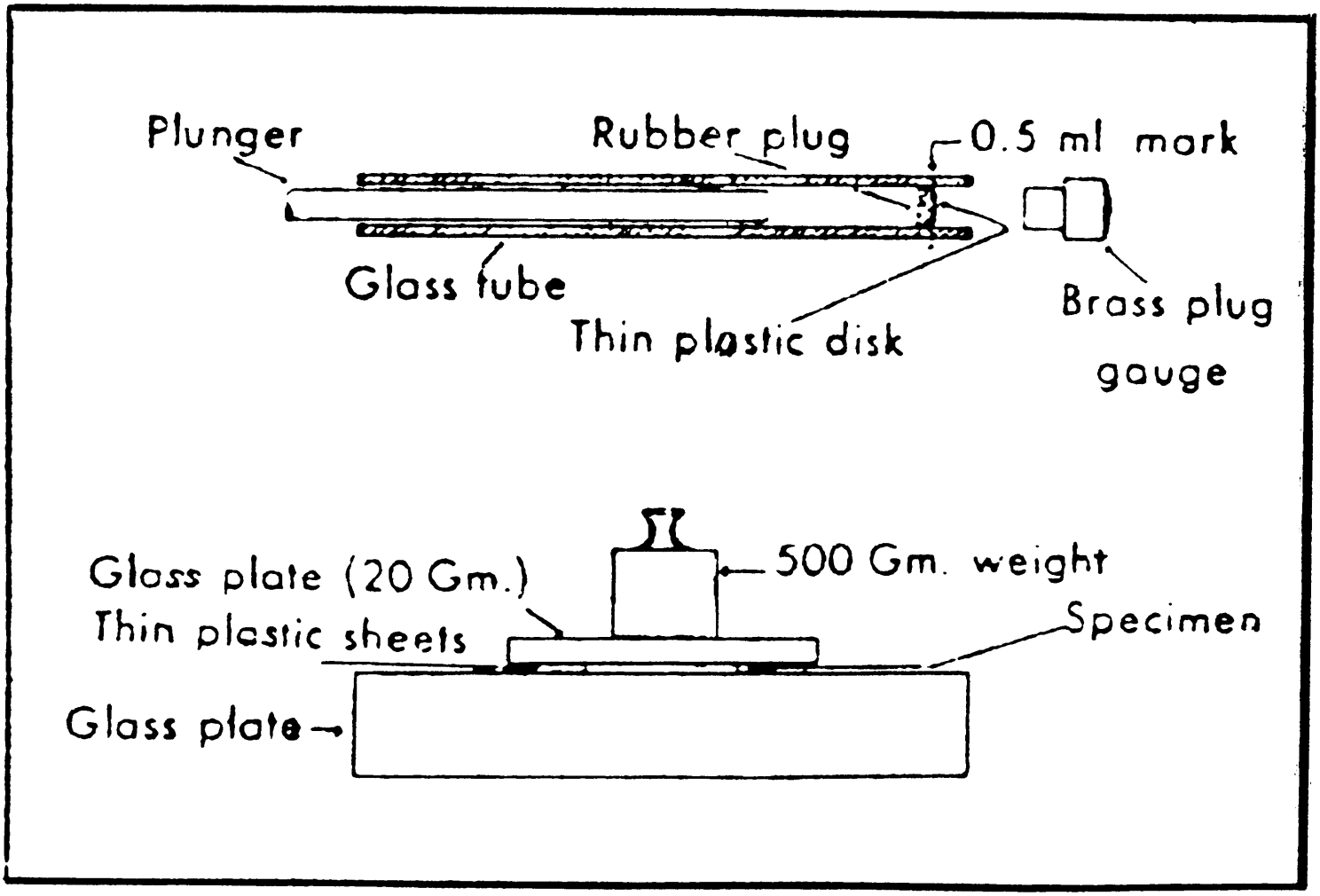


FIGURE 1. Apparatus for measuring the consistency of Zinc-Oxide Eugenol Impression Pastes

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4.4.3.3 Penetration hardness. The specimen used for measuring final set shall be used for measuring penetration hardness. The cup shall be placed on the platen of a Krebs penetrometer equipped with the standard penetrometer needle described in Method of Test for Penetration of Bituminous Materials, (ASTM) Specification D5 (2.2) and so loaded that the weight of the needle and load shall total 100 gm. Thirty minutes after mixing is commenced, the loaded needle shall be applied to the surface of the specimen for 10 seconds and the penetration recorded to the nearest 0.1 mm. The value for penetration shall be reported as the average of three determinations and shall be rounded off to the nearest 0.1 mm.

4.4.4 Separation from cast. A specimen shall be prepared as described in 4.4.3. The specimen shall be boxed with a strip of wax so that the wax extends 1 inch above the surface of the material. Five minutes after final setting, a mix of gypsum (dental stone), prepared according to the manufacturer's directions, shall be poured into the boxed specimen and vibrated to eliminate air bubbles at the interface. Two hours later, the specimen shall be placed in water at 60° C for 15 minutes. The cast shall then be separated from the impression and the surfaces in contact shall be examined visually for adherence. There shall be no evidence of adherence of the impression to the cast or of the cast to the impression.

4.5 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.2 Unit package. One tube of zinc-oxide paste, one tube of eugenol paste, two keys, and instructions for use, as specified, shall be packaged in a box conforming to PPP-B-566, or PPP-B-676. Box design shall include suitable interior fittings or shall contain sufficient cushioning material to insure against shifting or damage to the contents. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for dental impression material shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Forty-eight unit packages of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be as specified in the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification.

5.2.2 Level B. Forty-eight unit packages of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for dental impression material shall conform to MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Dental impression material packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Dental impression material packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Civil Agencies. Marking shall be as specified in Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Unit package. Each unit package shall be marked as specified in MIL-STD-129. Marking shall include the lot (control) no., contract or purchase order number and name of contractor and the legends: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)" and "DO NOT FREEZE" or similar commercial legends.

5.3.2.2 Levels A, B and C. Exterior container and unitized load shall be marked as specified in MIL-STD-129. Marking shall include the lot (control) no. and the legends: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)" and "DO NOT FREEZE" or similar commercial legends.

5.4 General.

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defect from its appearance or ... defects which

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-536-2770	IMPRESSION MATERIAL, DENTAL, Zinc-Oxide Eugenol Paste, Hard
6520-01-188-5420	IMPRESSION MATERIAL, DENTAL, Zinc-Oxide Eugenol Paste, Soft

6.3 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY:
VA-OSS

06805

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